

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 1 of 40

Topic: Palindrome Number

Definition: A palindrome reads the same from left to right and right to left. The value is checked by reversing it and comparing it with the original.

Question 1: Check whether a number is a palindrome.

Input

12321

Program

```
num = 12321
temp = num
rev = 0
while temp > 0:
    rev = rev * 10 + temp % 10
    temp //= 10
print(num == rev)
```

Output

True

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 2 of 40

Topic: Factorial Using Recursion

Definition: A factorial is the product of all positive integers up to a given number. Recursion solves it by reducing the problem to a smaller factorial.

Question 2: Find the factorial of a number using recursion.

Input

5

Program

```
def factorial(n):  
    if n == 0:  
        return 1  
    return n * factorial(n - 1)  
  
print(factorial(5))
```

Output

120

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 3 of 40

Topic: String Palindrome

Definition: A palindrome reads the same from left to right and right to left. The value is checked by reversing it and comparing it with the original.

Question 3: Check whether a string is a palindrome.

Input

```
"madam"
```

Program

```
text = "madam"  
print(text == text[::-1])
```

Output

```
True
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 4 of 40

Topic: Fibonacci Series

Definition: The Fibonacci series is a sequence where each term is the sum of the previous two terms. It commonly begins with 0 and 1.

Question 4: Print the Fibonacci series up to n terms.

Input

6

Program

```
n = 6
a, b = 0, 1
for _ in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

Output

0 1 1 2 3 5

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 5 of 40

Topic: Largest Element in a List

Definition: Finding the largest element identifies the greatest value in a collection. Each value can be compared while keeping track of the current maximum.

Question 5: Find the largest element in a list without using max().

Input

```
[23, 67, 12, 89, 45]
```

Program

```
nums = [23, 67, 12, 89, 45]
largest = nums[0]
for n in nums:
    if n > largest:
        largest = n
print(largest)
```

Output

```
89
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 6 of 40

Topic: List Element Frequency

Definition: Frequency counting records how many times each value occurs. A dictionary is commonly used to store each value and its count.

Question 6: Count the frequency of each element in a list.

Input

```
[1, 2, 2, 3, 1, 3, 3]
```

Program

```
nums = [1, 2, 2, 3, 1, 3, 3]
freq = {}
for n in nums:
    freq[n] = freq.get(n, 0) + 1
print(freq)
```

Output

```
{1: 2, 2: 2, 3: 3}
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 7 of 40

Topic: Armstrong Number

Definition: An Armstrong number equals the sum of its digits raised to the power of the number of digits. The calculated sum is compared with the original number.

Question 7: Check whether a number is an Armstrong number.

Input

153

Program

```
num = 153
digits = str(num)
total = sum(int(d) ** len(digits) for d in digits)
print(total == num)
```

Output

True

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 8 of 40

Topic: Remove Duplicates from a List

Definition: Remove Duplicates from a List is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 8: Remove duplicate elements from a list without using set().

Input

```
[1, 2, 2, 3, 4, 4, 5]
```

Program

```
nums = [1, 2, 2, 3, 4, 4, 5]
result = []
for n in nums:
    if n not in result:
        result.append(n)
print(result)
```

Output

```
[1, 2, 3, 4, 5]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 9 of 40

Topic: Second Largest Number

Definition: Finding the largest element identifies the greatest value in a collection. Each value can be compared while keeping track of the current maximum.

Question 9: Find the second largest number in a list.

Input

```
[34, 67, 12, 89, 45]
```

Program

```
nums = [34, 67, 12, 89, 45]
unique = list(set(nums))
unique.sort(reverse=True)
print(unique[1])
```

Output

```
67
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 10 of 40

Topic: Anagram Strings

Definition: Anagrams contain the same characters with the same frequencies in a different order. Sorting or character counting can be used to compare them.

Question 10: Check whether two strings are anagrams.

Input

```
"listen", "silent"
```

Program

```
s1 = "listen"  
s2 = "silent"  
print(sorted(s1) == sorted(s2))
```

Output

```
True
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 11 of 40

Topic: Vowels and Consonants

Definition: Vowels and Consonants is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 11: Count vowels and consonants in a string.

Input

```
"Python Programming"
```

Program

```
text = "Python Programming"
vowels = "aeiouAEIOU"
v = c = 0
for ch in text:
    if ch.isalpha():
        if ch in vowels:
            v += 1
        else:
            c += 1
print("Vowels:", v)
print("Consonants:", c)
```

Output

```
Vowels: 4
Consonants: 13
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 12 of 40

Topic: Reverse a String

Definition: Reverse a String is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 12: Reverse a string without using slicing.

Input

"Python"

Program

```
text = "Python"  
rev = ""  
for ch in text:  
    rev = ch + rev  
print(rev)
```

Output

nohtyP

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 13 of 40

Topic: Sum of Digits

Definition: Sum of Digits is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 13: Find the sum of digits of a number.

Input

12345

Program

```
num = 12345
total = 0
while num:
    total += num % 10
    num //= 10
print(total)
```

Output

15

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 14 of 40

Topic: Prime Number

Definition: A prime number is greater than 1 and has exactly two positive divisors. Primality can be checked by testing possible divisors.

Question 14: Check whether a number is prime.

Input

29

Program

```
num = 29
is_prime = num > 1
for i in range(2, int(num ** 0.5) + 1):
    if num % i == 0:
        is_prime = False
        break
print(is_prime)
```

Output

True

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 15 of 40

Topic: Prime Numbers in a Range

Definition: A prime number is greater than 1 and has exactly two positive divisors. Primality can be checked by testing possible divisors.

Question 15: Print all prime numbers in a given range.

Input

10 to 30

Program

```
start, end = 10, 30
for num in range(start, end + 1):
    if num < 2:
        continue
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            break
    else:
        print(num, end=" ")
```

Output

11 13 17 19 23 29

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 16 of 40

Topic: GCD of Two Numbers

Definition: The greatest common divisor is the largest positive integer that divides two numbers without a remainder. The Euclidean algorithm finds it efficiently.

Question 16: Find the GCD of two numbers.

Input

48, 18

Program

```
a, b = 48, 18
while b:
    a, b = b, a % b
print(a)
```

Output

6

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 17 of 40

Topic: LCM of Two Numbers

Definition: The least common multiple is the smallest positive number divisible by two given integers. It can be calculated using GCD and the numbers.

Question 17: Find the LCM of two numbers.

Input

12, 18

Program

```
a, b = 12, 18
x, y = a, b
while y:
    x, y = y, x % y
gcd = x
print(abs(a * b) // gcd)
```

Output

36

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 18 of 40

Topic: Word Count

Definition: Word Count is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 18: Count the number of words in a sentence.

Input

```
"Python is easy to learn"
```

Program

```
text = "Python is easy to learn"  
words = text.split()  
print(len(words))
```

Output

```
5
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 19 of 40

Topic: First Non-Repeating Character

Definition: First Non-Repeating Character is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 19: Find the first non-repeating character in a string.

Input

```
"swiss"
```

Program

```
text = "swiss"  
for ch in text:  
    if text.count(ch) == 1:  
        print(ch)  
        break
```

Output

```
w
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 20 of 40

Topic: Duplicate Elements in a List

Definition: Duplicate Elements in a List is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 20: Find duplicate elements in a list.

Input

```
[1, 2, 3, 2, 4, 3, 5]
```

Program

```
nums = [1, 2, 3, 2, 4, 3, 5]
seen = set()
duplicates = []
for n in nums:
    if n in seen and n not in duplicates:
        duplicates.append(n)
    seen.add(n)
print(duplicates)
```

Output

```
[2, 3]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 21 of 40

Topic: Sorting a List

Definition: Sorting a List is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 21: Sort a list without using sort().

Input

```
[5, 2, 9, 1, 5]
```

Program

```
nums = [5, 2, 9, 1, 5]
for i in range(len(nums)):
    for j in range(i + 1, len(nums)):
        if nums[i] > nums[j]:
            nums[i], nums[j] = nums[j], nums[i]
print(nums)
```

Output

```
[1, 2, 5, 5, 9]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 22 of 40

Topic: Missing Number

Definition: Missing Number is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 22: Find the missing number from 1 to n.

Input

```
[1, 2, 3, 5, 6]
```

Program

```
nums = [1, 2, 3, 5, 6]
n = 6
expected = n * (n + 1) // 2
print(expected - sum(nums))
```

Output

```
4
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 23 of 40

Topic: Move Zeros to the End

Definition: Move Zeros to the End is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 23: Move all zeros to the end of a list.

Input

```
[0, 1, 0, 3, 12]
```

Program

```
nums = [0, 1, 0, 3, 12]
non_zero = [n for n in nums if n != 0]
result = non_zero + [0] * (len(nums) - len(non_zero))
print(result)
```

Output

```
[1, 3, 12, 0, 0]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 24 of 40

Topic: Common Elements in Two Lists

Definition: Common Elements in Two Lists is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 24: Find common elements in two lists.

Input

```
[1, 2, 3, 4] and [3, 4, 5, 6]
```

Program

```
a = [1, 2, 3, 4]
b = [3, 4, 5, 6]
print([x for x in a if x in b])
```

Output

```
[3, 4]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 25 of 40

Topic: Merge Dictionaries

Definition: Merge Dictionaries is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 25: Merge two dictionaries.

Input

```
{"a": 1} and {"b": 2}
```

Program

```
d1 = {"a": 1}
d2 = {"b": 2}
merged = {**d1, **d2}
print(merged)
```

Output

```
{'a': 1, 'b': 2}
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 26 of 40

Topic: Maximum Value Key in a Dictionary

Definition: Maximum Value Key in a Dictionary is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 26: Find the key with the maximum value in a dictionary.

Input

```
{"a": 10, "b": 25, "c": 15}
```

Program

```
data = {"a": 10, "b": 25, "c": 15}  
print(max(data, key=data.get))
```

Output

```
b
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 27 of 40

Topic: Flatten a Nested List

Definition: Flatten a Nested List is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 27: Flatten a nested list.

Input

```
[[1, 2], [3, 4], [5]]
```

Program

```
nested = [[1, 2], [3, 4], [5]]
flat = []
for group in nested:
    flat.extend(group)
print(flat)
```

Output

```
[1, 2, 3, 4, 5]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 28 of 40

Topic: Set Intersection

Definition: Set Intersection is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 28: Find the intersection of two sets.

Input

```
{1, 2, 3} and {2, 3, 4}
```

Program

```
a = {1, 2, 3}
b = {2, 3, 4}
print(a & b)
```

Output

```
{2, 3}
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 29 of 40

Topic: Balanced Parentheses

Definition: Balanced Parentheses is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 29: Check balanced parentheses.

Input

```
"{[()]}"
```

Program

```
text = "{[()]}"
pairs = {")": "(", "]" : "[", "}": "{"
}
stack = []
for ch in text:
    if ch in "([{":
        stack.append(ch)
    elif ch in pairs:
        if not stack or stack.pop() != pairs[ch]:
            print(False)
            break
    else:
        print(not stack)
```

Output

```
True
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 30 of 40

Topic: Longest Word in a Sentence

Definition: Longest Word in a Sentence is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 30: Find the longest word in a sentence.

Input

```
"Python makes programming interesting"
```

Program

```
text = "Python makes programming interesting"  
words = text.split()  
print(max(words, key=len))
```

Output

```
programming
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 31 of 40

Topic: Remove Spaces from a String

Definition: Remove Spaces from a String is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 31: Remove all spaces from a string.

Input

```
"Python is fun"
```

Program

```
text = "Python is fun"  
print(text.replace(" ", ""))
```

Output

```
Pythonisfun
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 32 of 40

Topic: Character Frequency

Definition: Frequency counting records how many times each value occurs. A dictionary is commonly used to store each value and its count.

Question 32: Count character frequency in a string.

Input

```
"banana"
```

Program

```
text = "banana"
freq = {}
for ch in text:
    freq[ch] = freq.get(ch, 0) + 1
print(freq)
```

Output

```
{'b': 1, 'a': 3, 'n': 2}
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 33 of 40

Topic: Matrix Element Sum

Definition: Matrix Element Sum is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 33: Find the sum of elements in a matrix.

Input

```
[[1,2],[3,4]]
```

Program

```
matrix = [[1, 2], [3, 4]]  
total = sum(sum(row) for row in matrix)  
print(total)
```

Output

```
10
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 34 of 40

Topic: Matrix Transpose

Definition: Matrix Transpose is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 34: Transpose a matrix.

Input

```
[[1,2,3],[4,5,6]]
```

Program

```
matrix = [[1, 2, 3], [4, 5, 6]]
transpose = [list(row) for row in zip(*matrix)]
print(transpose)
```

Output

```
[[1, 4], [2, 5], [3, 6]]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 35 of 40

Topic: Stack Using a List

Definition: Stack Using a List is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 35: Implement a simple stack using a list.

Input

```
push 10, 20; pop once
```

Program

```
stack = []
stack.append(10)
stack.append(20)
print("Popped:", stack.pop())
print("Stack:", stack)
```

Output

```
Popped: 20
Stack: [10]
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 36 of 40

Topic: Binary Search

Definition: Binary search finds a target in a sorted list by repeatedly dividing the search range in half. Each comparison removes part of the search space.

Question 36: Implement binary search on a sorted list.

Input

```
[10, 20, 30, 40, 50], target=40
```

Program

```
nums = [10, 20, 30, 40, 50]
target = 40
low, high = 0, len(nums) - 1
found = False
while low <= high:
    mid = (low + high) // 2
    if nums[mid] == target:
        found = True
        break
    if nums[mid] < target:
        low = mid + 1
    else:
        high = mid - 1
print(found)
```

Output

```
True
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 37 of 40

Topic: Two Sum

Definition: Two Sum is a Python programming technique used to solve the given problem. It helps perform the required operation clearly and efficiently.

Question 37: Find the two numbers whose sum equals a target.

Input

```
[2, 7, 11, 15], target=9
```

Program

```
nums = [2, 7, 11, 15]
target = 9
for i in range(len(nums)):
    for j in range(i + 1, len(nums)):
        if nums[i] + nums[j] == target:
            print(nums[i], nums[j])
```

Output

```
2 7
```

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 38 of 40

Topic: Function Decorator

Definition: A decorator adds behavior around another function without changing its original code. It can run actions before or after the wrapped function.

Question 38: Use a decorator to print before and after a function call.

Input

No external input

Program

```
def log_call(func):
def wrapper():
print("Before")
func()
print("After")
return wrapper

@log_call
def greet():
print("Hello")

greet()
```

Output

Before
Hello
After

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 39 of 40

Topic: Even Number Generator

Definition: A generator produces values one at a time using yield instead of returning the complete sequence. This makes iteration memory-efficient.

Question 39: Create a generator that yields even numbers up to a limit.

Input

10

Program

```
def even_gen(limit):  
    for i in range(2, limit + 1, 2):  
        yield i  
  
print(list(even_gen(10)))
```

Output

[2, 4, 6, 8, 10]

TOP 40 PYTHON INTERVIEW QUESTIONS

Question 40 of 40

Topic: Division by Zero Exception Handling

Definition: Exception handling lets a program respond safely to runtime errors. ZeroDivisionError can be caught when division by zero occurs.

Question 40: Handle division by zero using exception handling.

Input

```
10 / 0
```

Program

```
try:
    print(10 / 0)
except ZeroDivisionError:
    print("Division by zero not allowed")
```

Output

```
Division by zero not allowed
```